

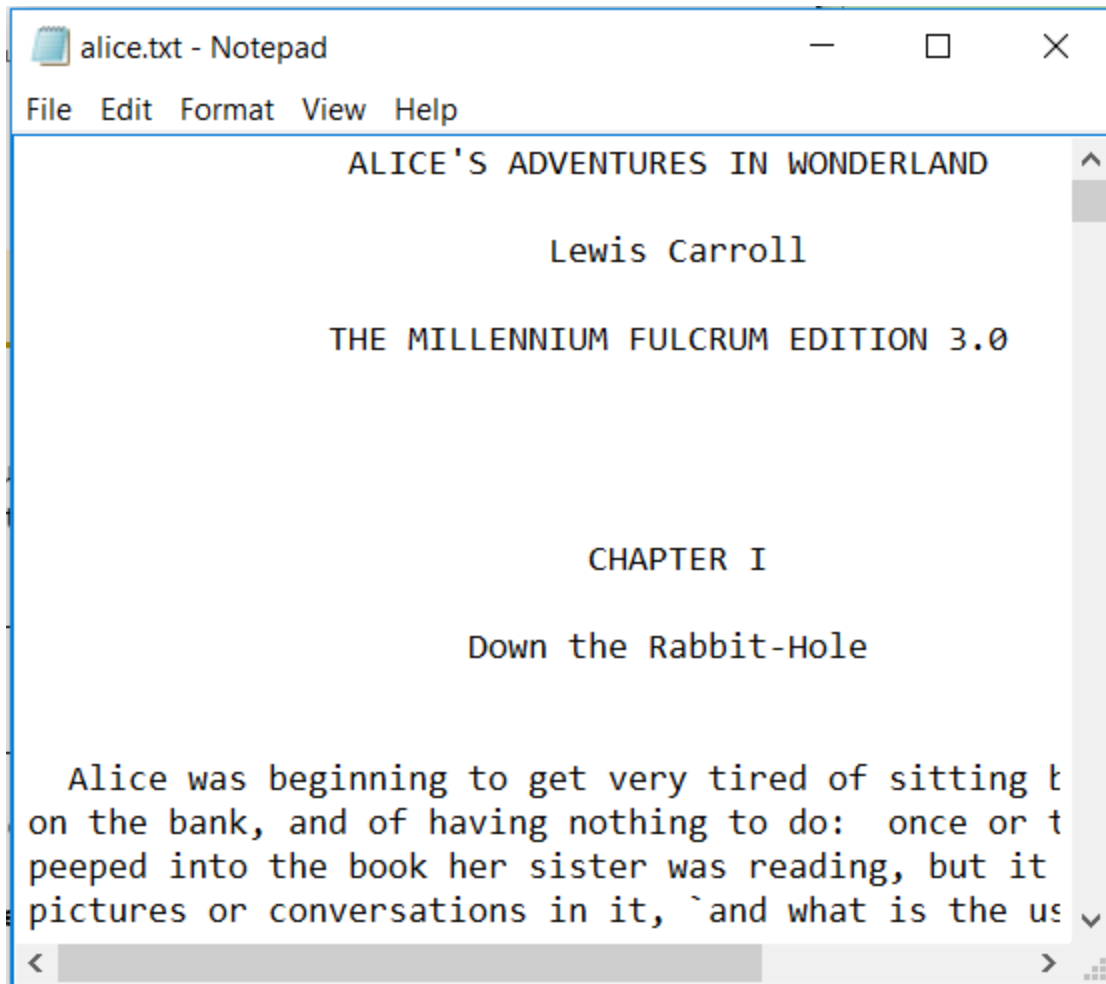
Worksheet

Computer Network - Week 4 - TCP

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Activity 1 : Sniffing a Big File Upload using TCP

Screenshots



*Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	EliteGro_80:40:d7	Broadcast	ARP	60	Who has 10.6.47.87? T
2	0.337748	10.6.47.43	74.125.68.95	UDP	71	64534 → 443 Len=29
3	0.366247	74.125.68.95	10.6.47.43	UDP	67	443 → 64534 Len=25
4	0.665871	EliteGro_80:40:d7	Broadcast	ARP	60	Who has 10.6.47.87? T
5	1.136950	10.6.47.43	128.119.245.12	TCP	54	49949 → 80 [FIN, ACK]
6	1.137122	10.6.47.43	128.119.245.12	TCP	66	49952 → 80 [SYN] Seq=
7	1.139028	10.6.47.43	128.119.245.12	TCP	779	49950 → 80 [PSH, ACK]
8	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
9	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
10	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
11	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=

> Frame 6: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{DE912EB7} ^

> Ethernet II, Src: EliteGro_30:2d:36 (94:c6:91:30:2d:36), Dst: 9e:e3:74:3e:cc:10 (9e:e3:74:3e:cc:10)

> Internet Protocol Version 4, Src: 10.6.47.43, Dst: 128.119.245.12

▼ Transmission Control Protocol, Src Port: 49952, Dst Port: 80, Seq: 0, Len: 0

Source Port: 49952
Destination Port: 80
[Stream index: 1]
[Conversation completeness: Incomplete, ESTABLISHED (7)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 593938918
[Next Sequence Number: 1 (relative sequence number)]

0000 9e e3 74 3e cc 10 94 c6 91 30 2d 36 08 00 45 00 ..t>....-0-6--E-
0010 00 34 49 96 40 00 80 06 02 79 0a 06 2f 2b 80 77 -4I-@-...-y-/+-w
0020 f5 0c c3 20 00 50 23 66 c9 e6 00 00 00 00 80 02 ...-P#f-.....
0030 20 00 ef 9d 00 00 02 04 05 b4 01 03 03 08 01 01
0040 04 02 ..

Transmission Control Protocol (tcp), 32 bytes | Packets: 181 · Displayed: 181 (100.0%) · Dropped: 0 (0.0%) | Profile: Default

*Ethernet

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No.	Time	Source	Destination	Protocol	Length	Info
13	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
14	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
15	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
16	1.139129	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
17	1.178143	10.6.47.43	74.125.68.95	UDP	71	64534 → 443 Len=29
18	1.205935	74.125.68.95	10.6.47.43	UDP	67	443 → 64534 Len=25
19	1.394424	128.119.245.12	10.6.47.43	TCP	60	80 → 49949 [ACK] Seq=
20	1.394763	128.119.245.12	10.6.47.43	TCP	66	80 → 49952 [SYN, ACK]
21	1.394875	10.6.47.43	128.119.245.12	TCP	54	49952 → 80 [ACK] Seq=
22	1.396598	128.119.245.12	10.6.47.43	TCP	60	80 → 49950 [ACK] Seq=
23	1.396672	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=

> Frame 20: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{DE912EB}

> Ethernet II, Src: 9e:e3:74:3e:cc:10 (9e:e3:74:3e:cc:10), Dst: EliteGro_30:2d:36 (94:c6:91:30:2d:36)

> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 10.6.47.43

▼ Transmission Control Protocol, Src Port: 80, Dst Port: 49952, Seq: 0, Ack: 1, Len: 0

Source Port: 80

Destination Port: 49952

[Stream index: 1]

[Conversation completeness: Incomplete, ESTABLISHED (7)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2460662571

[Next Sequence Number: 1 (relative sequence number)]

```

0000  94 c6 91 30 2d 36 9e e3 74 3e cc 10 08 00 45 00  --0-6-- t>---E-
0010  00 34 00 00 40 00 20 06 ac 0f 80 77 f5 0c 0a 06  -4-@- - -w-
0020  2f 2b 00 50 c3 20 92 aa bb 2b 23 66 c9 e7 80 12  /+P- - -+f-
0030  72 10 4f ab 00 00 02 04 05 b0 01 01 04 02 01 03  r-O-----
0040  03 07  --

```

Transmission Control Protocol (tcp), 32 bytes

Packets: 181 · Displayed: 181 (100.0%) · Dropped: 0 (0.0%) Profile: Default

*Ethernet

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Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
132	1.913655	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
133	2.168564	128.119.245.12	10.6.47.43	TCP	60	80 → 49950 [ACK] Seq=
134	2.168625	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
135	2.168625	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
136	2.168625	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [PSH, ACK] Seq=
137	2.168625	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
138	2.168625	10.6.47.43	128.119.245.12	TCP	1510	49950 → 80 [ACK] Seq=
139	2.168625	10.6.47.43	128.119.245.12	HTTP	917	POST /wireshark-labs/
140	2.169348	128.119.245.12	10.6.47.43	TCP	60	80 → 49950 [ACK] Seq=
141	2.169348	128.119.245.12	10.6.47.43	TCP	60	80 → 49950 [ACK] Seq=
142	2.169348	128.119.245.12	10.6.47.43	TCP	60	80 → 49950 [ACK] Seq=

> Frame 139: 917 bytes on wire (7336 bits), 917 bytes captured (7336 bits) on interface \Device\NPF_{DE...}

> Ethernet II, Src: EliteGro_30:2d:36 (94:c6:91:30:2d:36), Dst: 9e:e3:74:3e:cc:10 (9e:e3:74:3e:cc:10)

> Internet Protocol Version 4, Src: 10.6.47.43, Dst: 128.119.245.12

▼ Transmission Control Protocol, Src Port: 49950, Dst Port: 80, Seq: 152182, Ack: 1, Len: 863

Source Port: 49950

Destination Port: 80

[Stream index: 2]

[Conversation completeness: Incomplete (12)]

[TCP Segment Len: 863]

Sequence Number: 152182 (relative sequence number)

Sequence Number (raw): 1345155728

[Next Sequence Number: 153045 (relative sequence number)]

Offset	Hex	ASCII
0020	f5 0c c3 1e 00 50 50 2d 72 90 c3 dc a1 b5 50 18	...PP- r...P.
0030	01 05 d4 5b 00 00 20 61 6e 64 0d 0a 61 6c 6c 20	...[... a nd...all
0040	74 68 65 20 6f 74 68 65 72 20 71 75 65 65 72 20	the othe r queer
0050	6e 6f 69 73 65 73 2c 20 77 6f 75 6c 64 20 63 68	noises, would ch
0060	61 6e 67 65 20 28 73 68 65 20 6b 6e 65 77 29 20	ange (sh e knew)
0070	74 6f 20 74 68 65 0d 0a 63 6f 6e 66 75 73 65 64	to the.. confused
0080	20 63 6c 61 6d 6f 75 72 20 6f 66 20 74 68 65 20	clamour of the
0090	62 75 73 79 20 66 61 72 6d 2d 79 61 72 64 2d 2d	busy far m-yard--
00a0	77 68 69 6c 65 20 74 68 65 20 6c 6f 77 69 6e 67	while th e lowing
00b0	20 6f 66 20 74 68 65 0d 0a 63 61 74 74 6c 65 20	of the ..cattle
00c0	69 6e 20 74 68 65 20 64 69 73 74 61 6e 63 65 20	in the d istance
00d0	77 6f 75 6c 64 20 74 61 6b 65 20 74 68 65 20 70	would ta ke the p
00e0	6c 61 63 65 20 6f 66 20 74 68 65 20 4d 6f 63 6b	lace of the Mock

Frame (917 bytes) Reassembled TCP (153044 bytes)

Transmission Control Protocol (tcp), 20 bytes

Packets: 181 · Displayed: 181 (100.0%) · Dropped: 0 (0.0%) Profile: Default

Answers

1. What is the **sequence number** of the TCP **SYN** segment used to initiate the TCP connection between

the client computer and gaia.cs.umass.edu? What is contained in this TCP segment that **identifies** it as a SYN segment?

Sequence number: 593938918
In the Flags field, it identified as a SYN

2. What is the **sequence number** of the **SYNACK** segment sent by gaia.cs.umass.edu to the client computer in **reply to the SYN**? What is it in the segment that **identifies** the segment as a SYNACK segment? What is the value of the **Acknowledgement field** in the SYNACK segment? How did gaia.cs.umass.edu determine that **value**?

Sequence number: 2460662571
In the flags field, it identified as SYN, ACK, it indicate that the connection request is accepted
Acknowledgment number: 593938919
I think it comes from the SYN sequence number because it has the exact same values as the SYN segment. It happens because it accepts the connection request from the SYN.

3. What is the **sequence number** of the TCP segment containing the **header** of the **HTTP POST** command? How many **bytes of data** are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt **fit** into this single segment?

Sequence number: 1345155728
TCP Payload = 862 bytes
Yes it fit, you can see it in the line-based text data part in the MIME Multipart Media Encapsulation

4. What is the **length** (header and payload) of the segment that contains the **POST message header**?

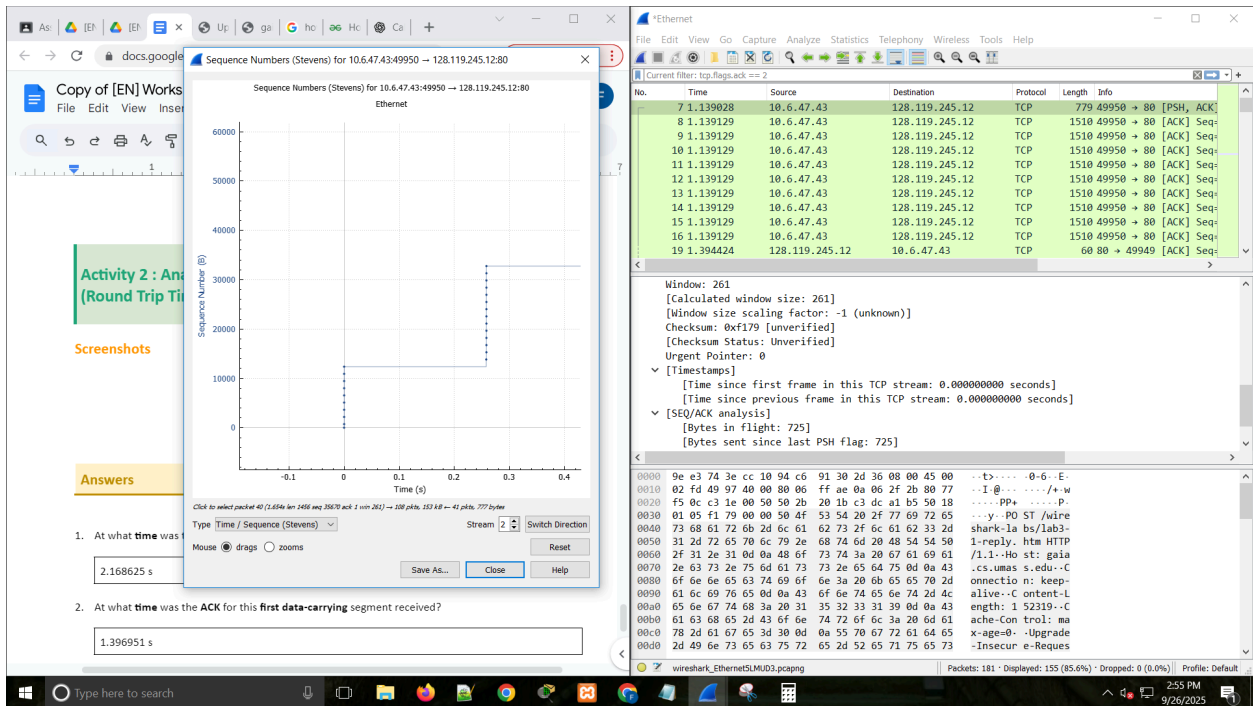
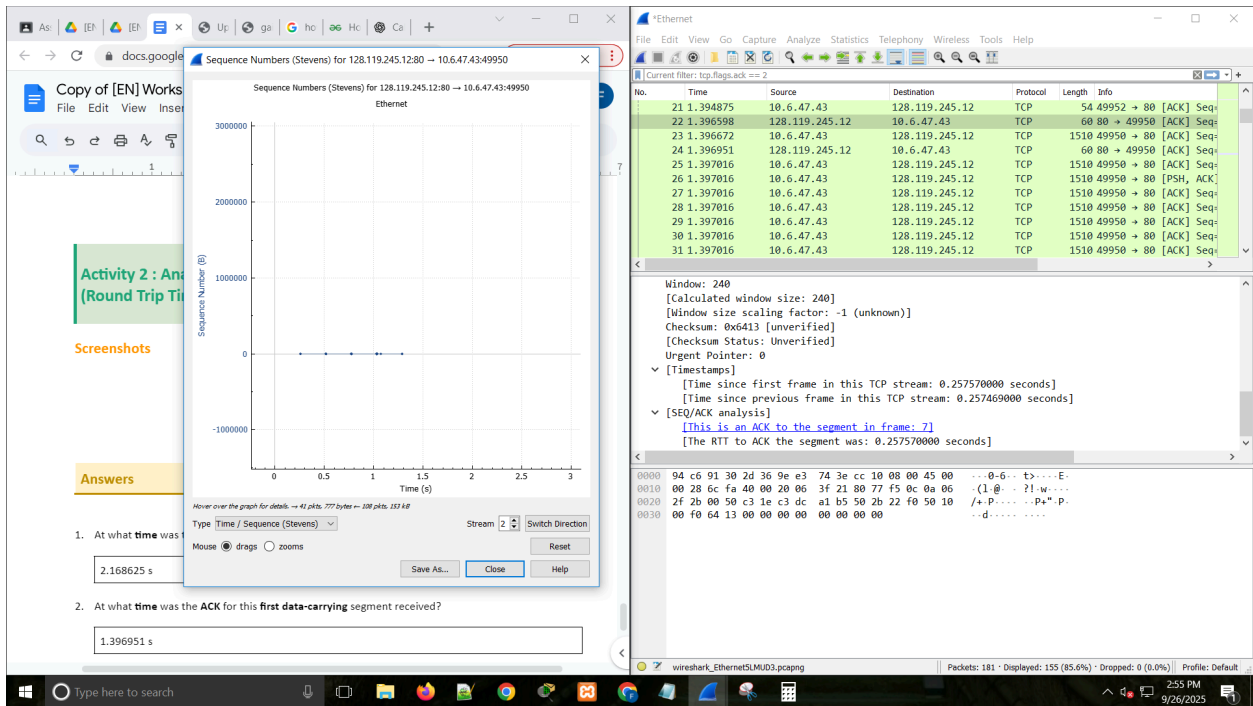
Header length: 20 bytes
Payload length: 862 bytes

5. How many **segments** were transmitted?

Segment count: 107, you can see it in Reassembled TCP segments

Activity 2 : Analyzing Wireshark Trace Files and Calculating RTT (Round Trip Time)

Screenshots



Activity 2 : Analysis (Round Trip Time)

Screenshots

Answers

- At what time was the first data-carrying segment received?
2.168625 s
- At what time was the ACK for this first data-carrying segment received?
1.396951 s

Sequence Numbers (Stevens) for 128.119.245.12:80 → 10.6.47.43:49950

Time (s)

Sequence Number (B)

Stream 2

Close

Window: 763

[Calculated window size: 763]

[Window size scaling factor: -1 (unknown)]

Checksum: 0xdf37 [unverified]

Checksum Status: Unverified

Urgent Pointer: 0

[Timestamps]

[Time since first frame in this TCP stream: 0.515657000 seconds]

[Time since previous frame in this TCP stream: 0.000000000 seconds]

[SEQ/ACK analysis]

[This is an ACK to the segment in frame: 37]

[The RTT to ACK the segment was: 0.257669000 seconds]

Activity 2 : Analysis (Round Trip Time)

Screenshots

Answers

- At what time was the first data-carrying segment received?
2.168625 s
- At what time was the ACK for this first data-carrying segment received?
1.396951 s

Sequence Numbers (Stevens) for 10.6.47.43:49950 → 128.119.245.12:80

Time (s)

Sequence Number (B)

Stream 2

Close

Window: 261

[Calculated window size: 261]

[Window size scaling factor: -1 (unknown)]

Checksum: 0x7ec5 [unverified]

Checksum Status: Unverified

Urgent Pointer: 0

[Timestamps]

[Time since first frame in this TCP stream: 0.257644000 seconds]

[Time since previous frame in this TCP stream: 0.000074000 seconds]

[SEQ/ACK analysis]

[Bytes in flight: 14560]

[Bytes sent since last PSH flag: 14560]

Answers

1. At what **time** was the **first segment** (the one containing the HTTP POST) sent?

2.168625 s

2. At what **time** was the **ACK** for this **first data-carrying** segment received?

1.396951 s

3. What is the **RTT** for the **first data-carrying** segment?

Receive - sent = 1.396951 - 1.178143 = 0.218808 s

4. What is the **RTT** for the TCP segment that carries the **second chunk** of data, and its **ACK**?

Receive - sent = 1.654685 - 1.396672 = 0.258013 s

5. What is the **Estimated RTT** after the **ACK** for the **second data** segment is received?

Estimated RTT = $(1-a) \cdot \text{estimatedRTT} + a \cdot \text{sampleRTT} = (1-0.125) \cdot 257.570 + 0.125 \cdot 257.669 = 258.161 \text{ ms}$